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Course	Digital Signal Processing Lab	**Experiment No.**	08
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Discrete Fourier Transform and its Inverse

The seasonal breathing of the planet in the Mauna Loa CO_2 record

Abstract: The monthly Mauna Loa CO_2 record (the Keeling Curve) was trimmed to 67 whole years (N = 804), detrended, and transformed with a DFT implemented from its definition, as a matrix product and by the FFT; the three agreed to 1e-13, the IDFT recovered the signal to 1e-12 ppm, and Parseval's theorem held to 1e-16. The spectrum consists of two clean lines at exactly 1 and 2 cycles per year (amplitudes 2.84 and 0.81 ppm) above a noise floor of about 0.016 ppm. An IDFT of these two lines alone rebuilt the seasonal cycle, which peaks in May and bottoms out in Oct (6.2 ppm peak to trough) and explains 88 % of the detrended variance; removing it reproduced NOAA's official deseasonalized series to 0.165 ppm. Cutting the record by 5 months caused leakage that lowered the annual line by 23 %, which a Hann window and zero padding largely corrected. The seasonal swing grew from 5.8 to 6.6 ppm over six decades.

Keywords: *DFT, IDFT, FFT, spectral leakage, windowing, zero padding, Parseval, frequency-domain filtering, Keeling Curve, CO_2, Python*

1. Aim and Objectives

Aim: to compute the DFT and IDFT of a real-world signal, verify their properties, and use them to analyse and filter the signal in the frequency domain, in Python.

- Implement the DFT and IDFT from the definition and as a matrix, and compare them with the FFT in accuracy and speed.
- Read frequencies, amplitudes and phases from the spectrum and understand leakage, windowing and zero padding.
- Use the IDFT to synthesise the seasonal cycle and to remove it, and validate the result against NOAA's own product.

2. Real-World Application

Atmospheric CO_2 has been measured continuously on Mauna Loa, Hawaii, since 1958. The record rises every year because of fossil-fuel burning, but it also 'breathes': each northern spring and summer, forests and crops take up CO_2, and each autumn and winter decaying vegetation releases it. Climate scientists separate this seasonal cycle from the long-term rise in exactly the way done here, and the DFT is the natural tool because the cycle is a sum of sinusoids at 1, 2, ... cycles per year. The DFT and IDFT also sit at the heart of many everyday technologies (Table 1).

Table 1. Where the DFT and IDFT are used

Technology	What the DFT / IDFT does
Climate science (this experiment)	separates seasonal cycles from trends; seasonal adjustment of indices
4G / 5G / Wi-Fi (OFDM)	the transmitter is an IDFT and the receiver a DFT, one per symbol
MRI scanners	the image is the 2-D IDFT of the measured k-space data
Audio and music apps	spectrum analysers, pitch detection, MP3 / AAC compression
Power-quality meters	measure harmonics of the 50 Hz mains (IEC 61000-4-7)

3. Theory

The **discrete Fourier transform** of an N-point sequence $x[n]$ and its **inverse** are

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}, \qquad x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\, e^{j2\pi kn/N} \tag{1}$$

for $k, n = 0, \ldots, N-1$. Collecting the twiddle factors $W_N^{kn} = e^{-j2\pi kn/N}$ in an $N \times N$ matrix gives the matrix form, whose inverse is its scaled conjugate:

$$\mathbf{X} = \mathbf{W}\mathbf{x}, \qquad \mathbf{W}^{-1} = \frac{1}{N}\mathbf{W}^{*} \tag{2}$$

Bin k corresponds to frequency $f_k = kf_s/N$, so the resolution is f_s/N. For a real signal a sinusoid $A\cos(2\pi k_0 n/N + \phi)$ appears in bins k_0 and $N - k_0$, from which its amplitude and phase are read:

$$f_k = \frac{kf_s}{N}, \qquad A = \frac{2|X[k_0]|}{N}, \qquad \phi = \angle X[k_0] \tag{3}$$

Two properties are checked numerically: energy is preserved (Parseval) and the spectrum of a real signal is conjugate-symmetric, so only bins 0 to N/2 carry information:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N}\sum_{k=0}^{N-1} |X[k]|^2, \qquad X[N-k] = X^{*}[k] \tag{4}$$

Leakage. The DFT implicitly repeats the N samples periodically. If a sinusoid does not complete a whole number of cycles in N samples, the repetition has a jump and the line spreads into neighbouring bins. A tapering window such as the Hann window reduces the spread, and zero padding to PN points samples the same spectrum more finely without adding resolution:

$$w[n] = \frac{1}{2} - \frac{1}{2}\cos\frac{2\pi n}{N}, \qquad X_{pad}[k] = \sum_{n=0}^{N-1} x[n]\, w[n]\, e^{-j2\pi kn/(PN)} \tag{5}$$

> **Note.** Direct evaluation of Eq. 1 needs N^2 complex multiplications; the fast Fourier transform (FFT) computes exactly the same numbers in about $N \log_2 N$ by reusing partial sums. It is fastest when N has only small prime factors.

4. Dataset Description

The **Mauna Loa monthly mean CO_2** record is published by the NOAA Global Monitoring Laboratory (in cooperation with Scripps Institution of Oceanography, which began it under C. D. Keeling). Values are dry-air mole fractions in parts per million (ppm). The CSV is 38 kB, freely downloadable and cached. It runs from 1958-03 to 2026-09 (823 months); NOAA also supplies a deseasonalized column, used here only as an independent check. For the DFT the record is trimmed to 1959-01 to 2025-12: N = 804 months = 67 whole years, so that 1 cycle per year falls exactly on bin k = 67.

Table 2. Dataset attributes

Property	Value	Property	Value
Provider	NOAA GML / Scripps	Station	Mauna Loa Observatory, Hawaii (3,397 m)
Full record	1958-03 to 2026-09	DFT record	1959-01 to 2025-12, N = 804
Range	312.42 to 432.34 ppm	Sampling	fs = 12 per year (monthly)
Resolution fs/N	0.0149 cycles/year	Nyquist frequency	6 cycles per year
Columns used	year, month, average, deseasonalized	Months without day count	195 (early record)

Table 3. Last six samples of the DFT record

Year	Month	Decimal year	CO_2 (ppm)	NOAA deseasonalized	Trend	Detrended r[n]
2025	7	2025.5417	427.87	427.49	426.73	+1.14
2025	8	2025.6250	425.48	427.41	426.95	-1.47
2025	9	2025.7083	424.37	427.85	427.18	-2.81
2025	10	2025.7917	424.87	428.12	427.40	-2.53
2025	11	2025.8750	426.46	428.40	427.62	-1.16
2025	12	2025.9583	427.49	428.22	427.85	-0.36

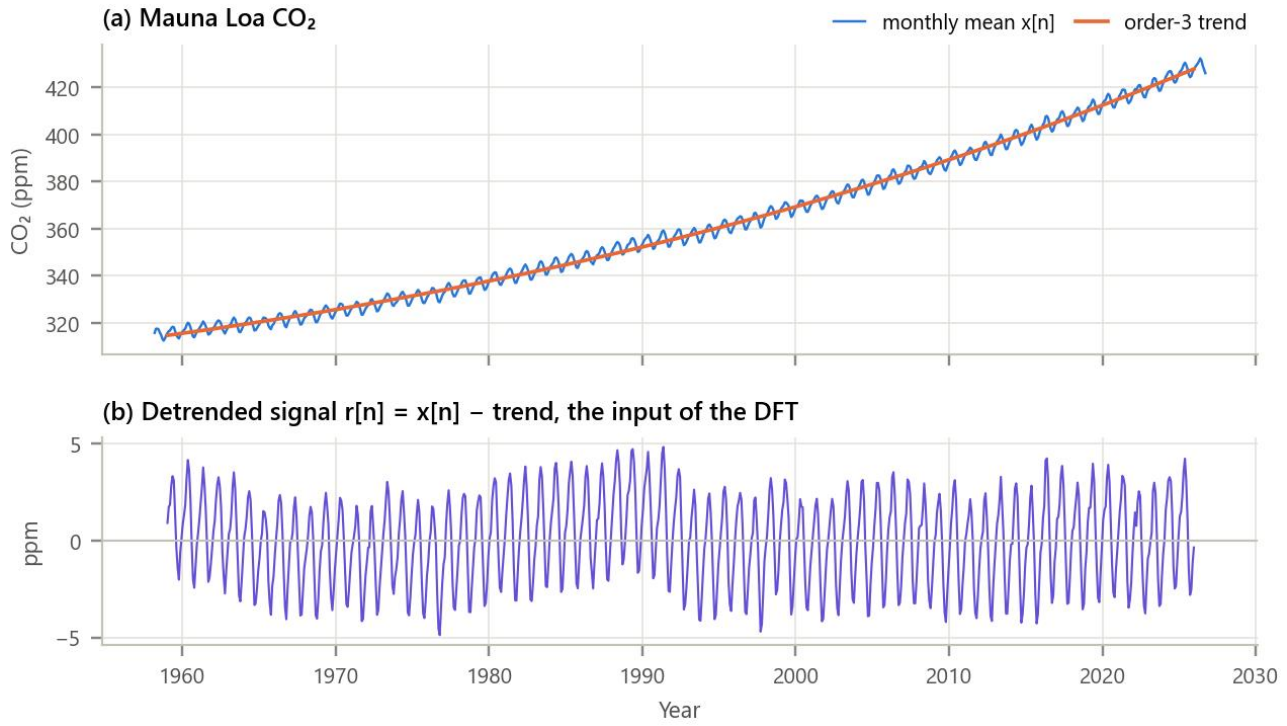


Figure 1. The input signal. (a) Monthly CO_2 with the order-3 polynomial trend; (b) the detrended signal.

Inference. CO_2 rose from about 316 to 426 ppm, and the rise is accelerating (the slope of the trend grew from 0.96 ppm/yr around 1960 to 2.57 ppm/yr around 2020). After the trend is removed, what remains is a strikingly regular oscillation of about ±3 ppm every year: a periodic signal for which the DFT is the natural description.

5. Methodology and Implementation

The trend is removed by least squares, the DFT of r[n] is computed, the bins at 1 and 2 cycles per year (k = 67 and 134, with their mirrors) are kept and all others set to zero, and the IDFT of what remains is the seasonal model. Leakage is studied by deleting 5 months, and the DFT matrix is timed against the FFT for several N.

Table 4. Software modules

Module	Responsibility
`config.py`	URL, sampling rate, trend order, harmonics, leakage and timing settings
`data_loader.py`	Download once, cache, trim to whole years
`dft.py`	DFT by definition, DFT matrix, IDFT, amplitude/phase, bin masking, Hann window, checks
`analysis.py`	Spectrum, leakage, seasonal synthesis, decades, growth rate, timing
`plots.py` / `gui.py`	Report figures / interactive spectrum analyser and DFT calculator
`test_dft.py`	10 unit tests (inverse, Parseval, symmetry, linearity, shift, leakage): all pass

Core of dft.py

```python
def dft_direct(x):                          # X[k] = sum x[n] e^{-j 2 pi k n / N}
    N = len(x)
    return np.array([sum(x[n] * np.exp(-2j * np.pi * k * n / N) for n in range(N))
                     for k in range(N)])

def dft_matrix(N):                          # W[k, n] = e^{-j 2 pi k n / N}
    k = np.arange(N)
    return np.exp(-2j * np.pi * np.outer(k, k) / N)

def idft(X):                                # x = (1/N) W* X
    return np.conj(dft_matrix(len(X))) @ X / len(X)

def keep_bins(X, bins):                     # ideal filter: keep +k and N-k only
    mask = np.zeros(len(X), bool)
    for k in bins:
        mask[k % len(X)] = mask[-k % len(X)] = True
    return np.where(mask, X, 0)

```

GUI. `gui.py` (Figure 2) lets the user change the record length month by month, the trend order, the window and the zero-padding factor, and choose which harmonics the IDFT keeps. It reports the line amplitude, whether N/12 falls on a bin, the resolution and the error against NOAA. A calculator returns the DFT of any typed sequence for checking hand calculations. Run `python run_experiment.py` and `python gui.py`.

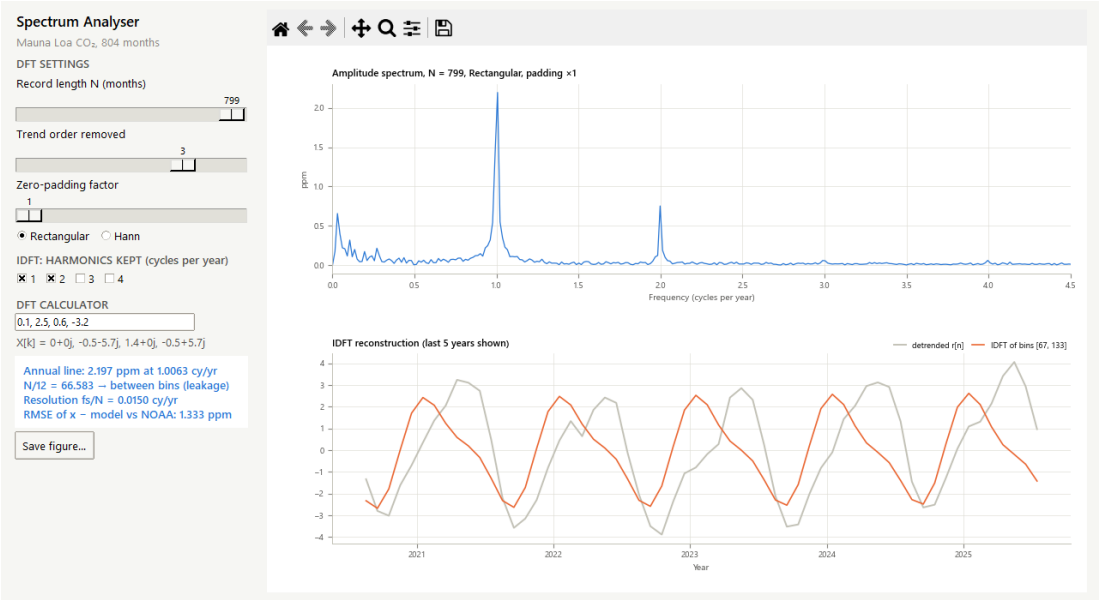


Figure 2. The spectrum analyser with N = 799 months. N/12 is not an integer: the annual line leaks, and an IDFT that keeps the nearest bins drifts out of phase.

6. Calculations

4-point DFT by hand. One sample per quarter of the seasonal model (January, April, July, October), rounded: x = [0.1, 2.5, 0.6, −3.2]. With N = 4, $W_4 = e^{-j\pi/2} = -j$, so W_4^{kn} cycles through 1, −j, −1, j:

Table 5. DFT of four quarterly values

k	X[k] = x[0] + x[1]W^{4k} + x[2]W^{42k} + x[3]W^{43k}	X[k]	|X[k]|	2|X[k]|/N
0	0.1 + 2.5 + 0.6 − 3.2	0.00	0.00	n/a (mean)
1	0.1 + 2.5(−j) + 0.6(−1) − 3.2(j)	-0.50 − j5.70	5.72	2.86
2	0.1 − 2.5 + 0.6 + 3.2	1.40	1.40	0.70
3	0.1 + 2.5(j) + 0.6(−1) − 3.2(−j)	-0.50 + j5.70	5.72	mirror of k = 1

Checks. X[3] is the complex conjugate of X[1] (symmetry). Parseval: Σx² = 16.86 and (1/4)Σ|X|² = (0 + 32.74 + 1.96 + 32.74)/4 = 16.86. IDFT of x[1]: (1/4)[X[0] + X[1]·j + X[2]·(−1) + X[3]·(−j)] = (1/4)(0 + 5.7 − 0.5j − 1.4 + 5.7 + 0.5j) = 2.5. **Real record:** |X[67]| = 1142.8, so A = 2|X|/N = 2.843 ppm with phase φ = -1.665 rad. The fundamental alone peaks where 2πn/12 + φ = 0, i.e. n = 3.18 months after the start of January (mid-April); adding the 2-cycle harmonic moves the combined peak to May. These values match the program (checked by `test_dft.py`).

7. Results and Discussion

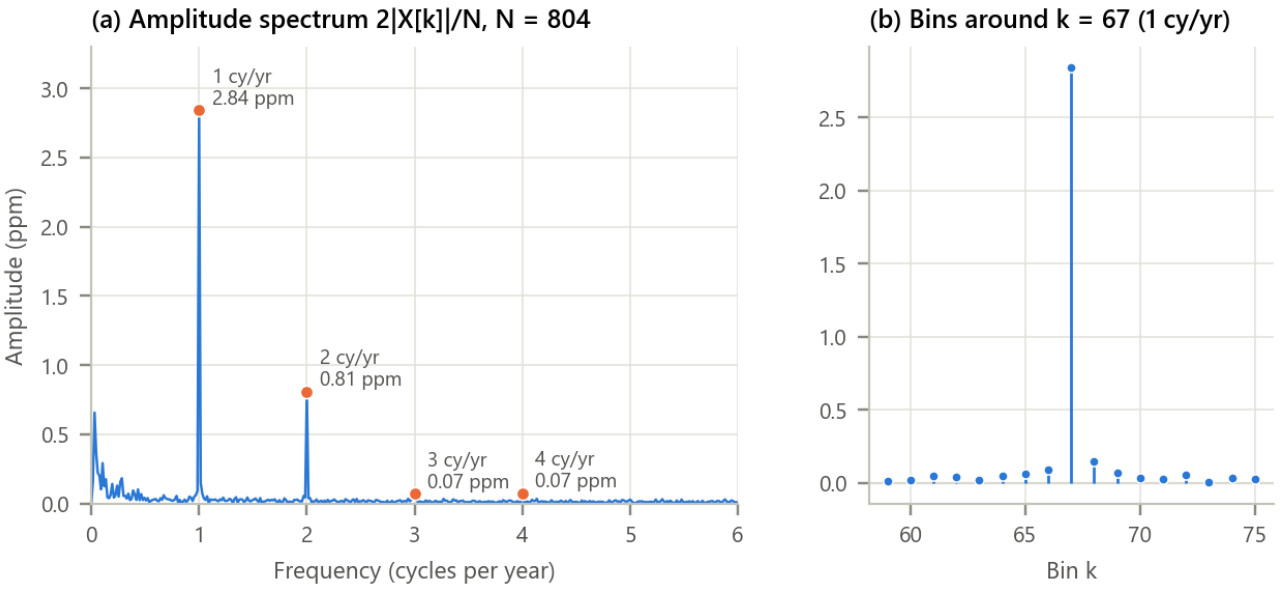


Figure 3. Amplitude spectrum of the detrended record. (a) 0 to 6 cycles per year (Nyquist); (b) individual bins around k = 67.

Inference.

- The spectrum is almost empty except for two lines exactly at 1 and 2 cycles per year: 2.843 ppm and 0.807 ppm, about 177 and 50 times the noise floor (0.016 ppm). The 3 and 4 cycle harmonics are tiny (0.068 and 0.072 ppm), so the seasonal cycle is well described by two sinusoids.
- Because N = 804 is a whole number of years, the annual line falls on a single bin: its neighbours are at noise level, i.e. there is no leakage. The small bump near 0 cycles per year is slow variation in the growth rate that a cubic trend does not capture.

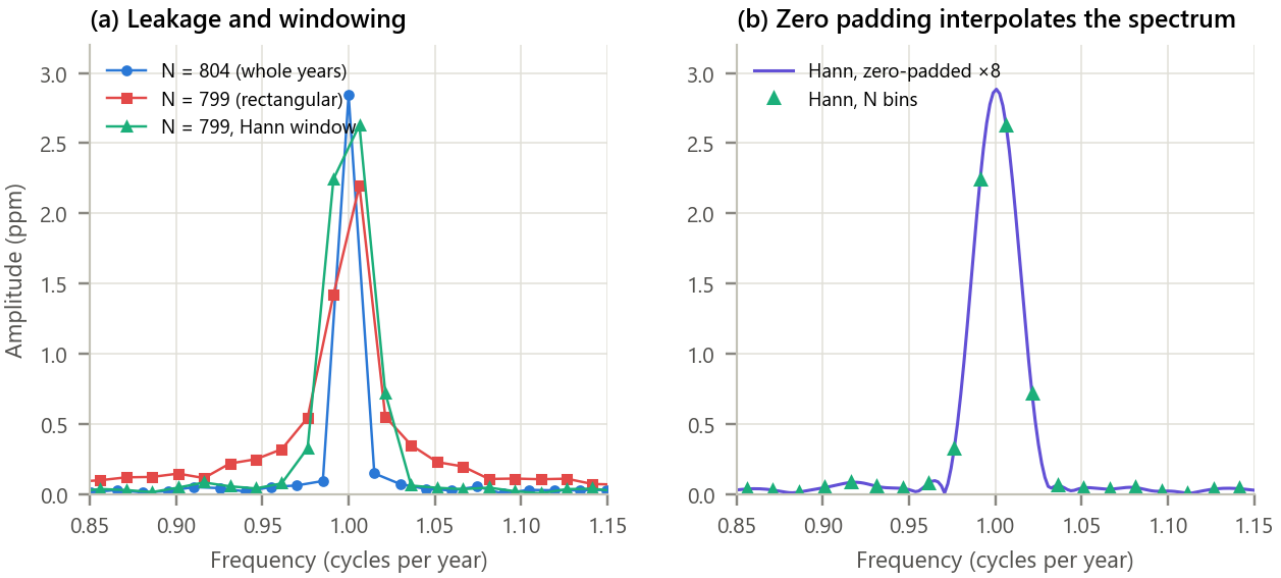


Figure 4. Leakage, windowing and zero padding. (a) N = 804 versus N = 799 with rectangular and Hann windows; (b) Hann-windowed spectrum zero-padded ×8.

Inference. With 799 months, N/12 is not an integer (the annual line sits 0.58 of a bin between bins), and the line leaks: its peak drops from 2.84 to 2.20 ppm and its energy spreads into neighbouring bins. The Hann window recovers the peak to 2.63 ppm and suppresses the far leakage, at the cost of a wider main lobe. Zero padding then shows the underlying curve: its maximum is 2.88 ppm at 1.0006 cycles per year, close to the true values. Zero padding interpolates; it does not sharpen the line.

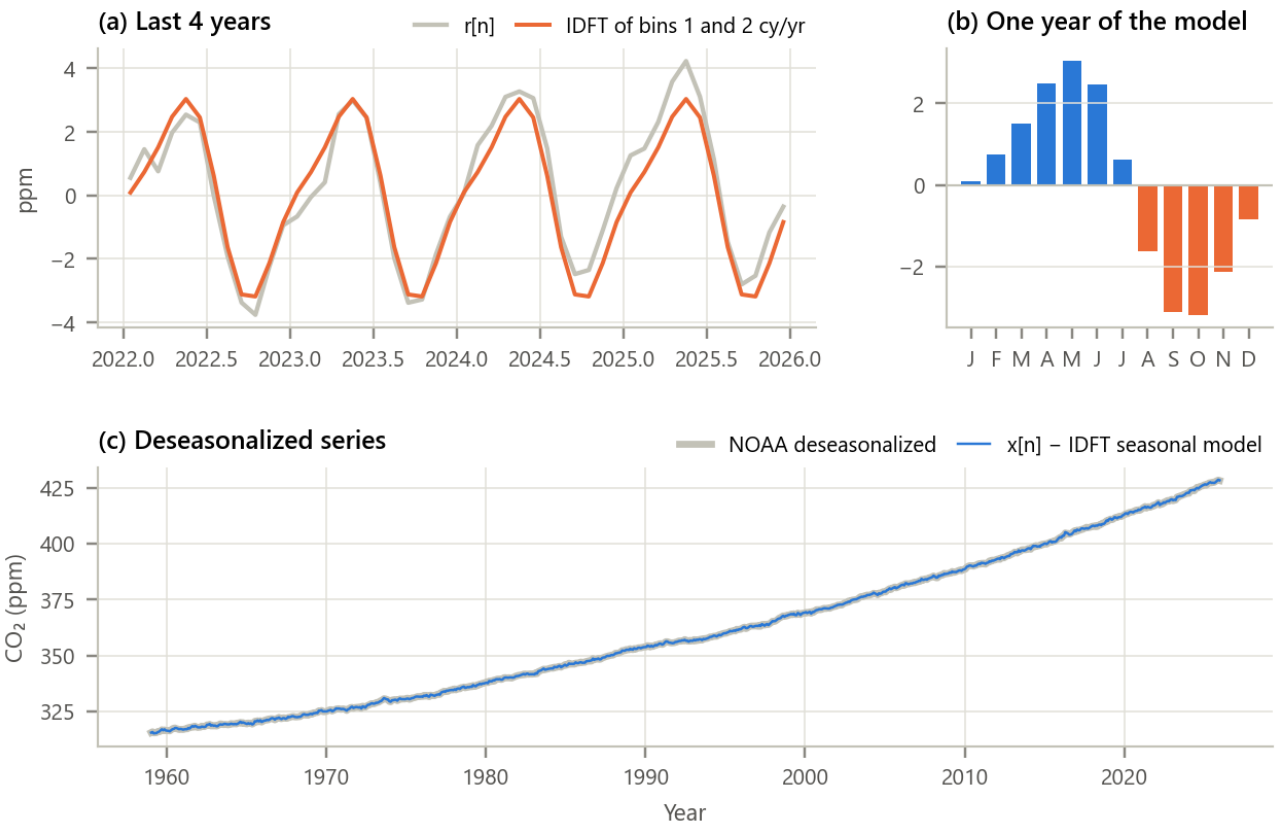


Figure 5. Synthesis and removal of the seasonal cycle with the IDFT. (a) Detrended signal and the IDFT of the two seasonal lines; (b) one year of the model; (c) deseasonalized CO_2 compared with NOAA's product (RMSE 0.165 ppm).

Inference.

- Keeping only 4 of 804 DFT coefficients reproduces the seasonal oscillation and explains 88 % of the detrended variance; what is left (std 0.76 ppm) is weather-driven noise and year-to-year growth variation.
- CO_2 is highest in **May** and lowest in **Oct**, a 6.2 ppm swing: the Northern Hemisphere biosphere draws CO_2 down through the growing season and releases it in autumn and winter.
- Subtracting the model gives a deseasonalized curve within 0.165 ppm of NOAA's official series, which is produced by a different method, so the simple DFT/IDFT filter is fit for purpose.

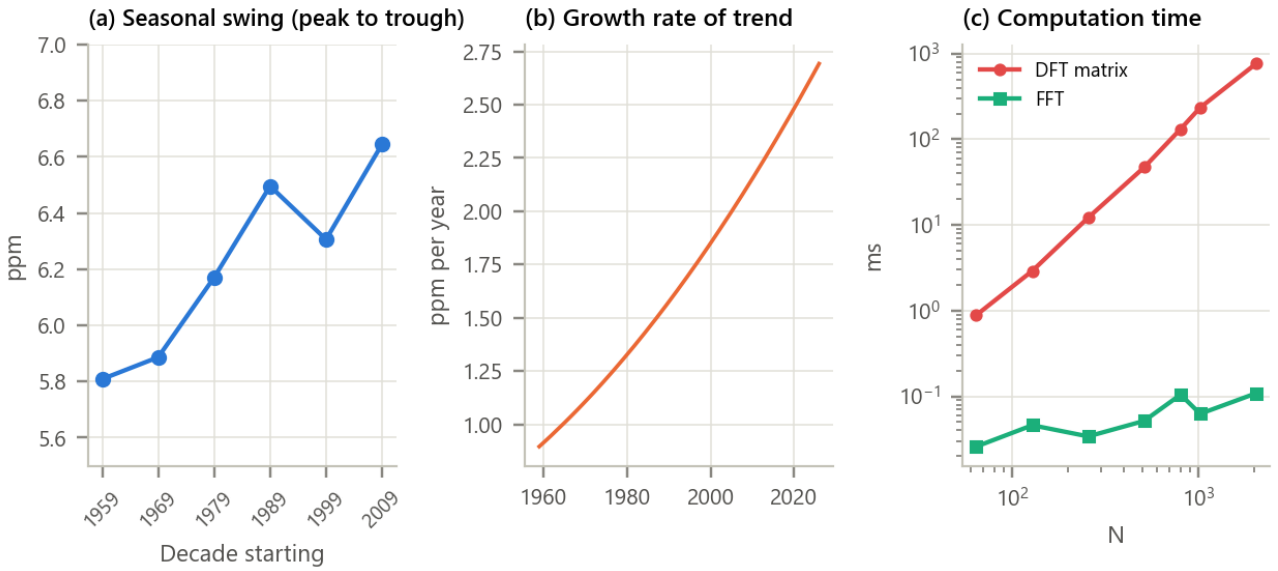


Figure 6. Long-term changes and computational cost. (a) Peak-to-trough seasonal swing per decade (IDFT of two lines); (b) growth rate of the trend; (c) time for one transform.

Inference.

- The seasonal swing grew from 5.81, 5.89, 6.17, 6.50, 6.31, 6.64 ppm decade by decade, about 14 % in total, consistent with published reports of a strengthening seasonal exchange in the north. Meanwhile the growth rate of CO_2 itself nearly tripled.
- The DFT matrix takes 120 ms for N = 804 and grows as N^2; the FFT takes 0.09 ms, about 1271 times faster. The FFT is slower for N = 804 than for N = 1024 (0.06 ms) because 804 = 4·3·67 has a large prime factor.

Table 6. Summary of results

Quantity	Result
DFT matrix / definition versus FFT	1e-13 / 4e-14
IDFT reconstruction error; Parseval; symmetry	1e-12 ppm; 1e-16; 1e-16
Annual / semi-annual line	2.843 / 0.807 ppm at 1.000 / 2.000 cy/yr
Seasonal maximum, minimum, swing	May, Oct, 6.22 ppm
Deseasonalized versus NOAA (RMSE)	0.165 ppm
Leakage (N = 799): rectangular / Hann / Hann + padding	2.20 / 2.63 / 2.88 ppm
Seasonal swing 1959-1968 / 2009-2018	5.81 / 6.64 ppm
FFT speed-up at N = 804	1271 ×

8. Conclusion

The DFT and IDFT were implemented from first principles and verified against the FFT and against their defining properties to machine precision. Applied to the Keeling Curve, the DFT reduced 67 years of monthly data to two numbers that matter, the 2.84 ppm annual and 0.81 ppm semi-annual lines, and the IDFT of those lines alone rebuilt the planet's seasonal breathing (maximum in May, minimum in Oct) well enough to reproduce NOAA's deseasonalized record within 0.17 ppm. The experiment also exposed the practical side of the transform: choose N to contain whole periods or use a window to control leakage,

remember that zero padding interpolates but does not resolve, and use the FFT, ideally with a highly composite N, because the direct sum quickly becomes too slow.

9. Learning Outcomes

- Compute the DFT and IDFT by definition, as a matrix product and with the FFT.
- Convert bins to frequencies and read amplitudes and phases of sinusoidal components.
- Explain and control spectral leakage with record length, windowing and zero padding.
- Filter in the frequency domain by editing DFT coefficients and inverting with the IDFT.

10. References

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